Rohan Kumar

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Education

University of Illinois at Urbana-Champaign **Expected Graduation Date: December 2026**
B.S. Computer Science and Statistics, GPA: 3.86 / 4.00 *Urbana-Champaign Area, IL*
Courses: Object Oriented Programming, Data Structures, Algorithms, Computer Systems, Database Systems, Distributed Systems, High Frequency Trading Technology, Statistical Modeling, Statistical Learning, Applied Machine Learning

Skills

Programming Languages: Python, TypeScript, JavaScript, Java, C, C++, C#, SQL
Web, Backend & Data: React, HTML/CSS, Node.js, FastAPI, .NET, PostgreSQL, Spark/PySpark, REST APIs, Playwright
Cloud & Infrastructure: AWS, GCP, Docker, Kubernetes, Linux, Cloudflare, GitHub Actions, CI/CD, Distributed Systems
AI Systems & Developer Tools: LangGraph, LangChain, RAG, LLM Evaluation, Claude Code, Codex, Cursor

Experience

Founding Engineer August 2026 – Present
Vinskal *Urbana-Champaign Area, IL*
- Architected the continuous job-discovery pipeline behind 500K+ live job postings using PostgreSQL lease claims and per-host rate pacing; diagnosed a 50-hour, 49K+ row silent stall and redesigned the queue for durable recovery.
- Built an agentic incident responder (Cloudflare Workers, GitHub Actions) and its production alert-routing middleware, autonomously investigating downtime, quota, and runtime failures and opening PRs for human review (1K+ alerts handled).
- Built a Playwright-based browser job application agent supporting five ATS platforms and validated on live employer submissions; developed an evaluation harness measuring reliability, model quality, and per-run cost.
- **Technologies:** Python, TypeScript, FastAPI, PostgreSQL, Playwright, Docker, Cloudflare Workers, GitHub Actions

Software Engineer Intern May 2026 – August 2026
Pinterest *New York, NY*
- Built a multi-agent AI system (TypeScript, LangGraph) that autonomously searches Pinterest's source code for security vulnerabilities, grounded in a retrieval (RAG) layer over the company's own history of confirmed vulnerabilities.
- Engineered a second agentic harness that dynamically validates suspected vulnerabilities by driving a Chrome browser, logging into real accounts, attempting to perform the claimed attack, and returning a Reproduced/Refuted verdict.
- Owned the platform end-to-end as the sole engineer building the infrastructure beneath both systems: a fleet-wide rate governor and a Postgres-backed job queue that keep hours-long agent runs durable under a shared LLM token budget.
- **Technologies:** TypeScript, Python, PostgreSQL, Prisma, LangGraph, DeepAgents, RAG, LLMs, Docker, REST APIs, Git

Software Development Engineer Intern February 2026 – April 2026
Amazon *Seattle, WA*
- Designed and deployed multilingual personalization features for Amazon Music's voice recommendation system (≈23M+ daily requests), integrating user listening behavior into an ML ranking pipeline and achieving 96%+ feature coverage.
- Owned end-to-end system design and development of a language-aware candidate filtering system in Java, reducing irrelevant cross-language recommendations and improving music recommendation quality across 17M+ daily voice requests.
- Engineered 5 large-scale PySpark data pipelines (AWS Glue) to analyze 100M+ recommendation events, uncovering feature coverage gaps and critical quality issues that directly informed ranking model inputs and system design decisions.
- **Technologies:** Java, PySpark, AWS (Glue), Reinforcement Learning (RL), A/B Testing, Distributed Systems

Software Engineer Intern May 2025 – August 2025
Relativity *Chicago, IL*
- Owned the end-to-end development of internal API extensions and automated GitHub Actions CI/CD workflows, reducing direct client data access during incident resolution by 20%+ and improving resolution speed.
- Designed and engineered fault-tolerant .NET (C#) migration jobs to transition over 1TB of data from legacy SQL systems to a distributed, Azure-hosted NoSQL architecture, enhancing horizontal scalability and reducing storage costs.
- **Technologies:** C#, .NET, Azure (AKS), Kubernetes, Docker, MySQL, NoSQL, REST APIs, GitHub Actions

Projects

Market Data Feeds Compression – FinTech Lab @ UIUC
- Designed and implemented a high-performance C++17 market data system for IEX TOPS/DEEP and NASDAQ ITCH PCAP feeds, using Zstandard (Zstd) with dictionary training to reduce data size by 80-85% (≈15% gain over generic Zstd).
- Engineered a streaming, memory-efficient architecture ($\leq$ 256MB) capable of processing multi-GB captures at 40K+ packets/sec, suitable for continuous market-hours ingestion while maintaining fast decompression and system reliability.
- **Technologies:** C++17, libpcap, Zstandard (Zstd), Make, Linux, Data Infrastructure, High Frequency Trading (HFT)

Publications

RIPEL: A Data-Augmented Peer Evaluation System for Assessing Teamwork February 2026
ACM SIGCSE TS 2026 (Poster)
- Wenxuan Shi, Jiaqi Linna Niu, Rohan Kumar, Yifan Song, Brian Bailey · pp. 1517–1518 · DOI: 10.1145/3770761.3777297